Statistical Inference for USD/SGD Exchange-Rate Returns

CLT, Bootstrap, and Sign-Flip Permutation Methods

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Research Question

Inference question

Does the mean of the observed USD/SGD log returns differ from zero, and do CLT/Wald and resampling-based confidence intervals and tests lead to consistent conclusions?

- Exchange-rate movements affect firms with cross-border trade and investors holding currency exposure.
- We focus on average return and its statistical uncertainty.
- This is not a direct test of MAS policy effectiveness or market efficiency.

Data and Measurement

- Source: NBER Macrohistory Database, series `DEXSIUS`.
- Sample: 27 Sep 2021 – 25 Sep 2026.
- 1,305 dated source rows; 55 have no exchange-rate value.
- 1,250 observed rates produce 1,249 non-missing returns.

Return definition

$$r_t = 100 \ln\left(\frac{S_t}{S_{t-1}}\right)$$

where S_t is the exchange-rate level reported by the source.

The first return is missing because there is no preceding observation. Some retained observations are more than one calendar day apart.

Methods

Estimate and quantify uncertainty

- CLT/Wald 95% interval: $\bar{r} \pm 1.96\, s/\sqrt{n}$
- Percentile bootstrap 95% interval
- Studentized bootstrap 95% interval

Test for zero mean

$$H_0 : \mu = 0 \qquad \text{vs.} \qquad H_1 : \mu \neq 0$$

- Wald z-test using $s/\sqrt{n}$
- Sign-flip permutation test

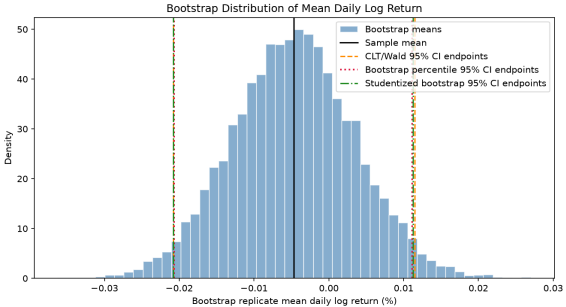
Each resampling method uses $B = 10{,}000$ replicates and random seed 60702. The sign-flip test assumes symmetry under the null.

Confidence Intervals: Similar Conclusions

Estimated mean: $\bar{r} = -0.004664\%$

Method	Lower	Upper
CLT/Wald	-0.020819	0.011491
Percentile bootstrap	-0.020702	0.011136
Studentized bootstrap	-0.020769	0.011241

- All intervals include zero.
- CLT SE: 0.008242%; bootstrap SE: 0.008196%.



Bootstrap distribution of the sample mean and interval endpoints.

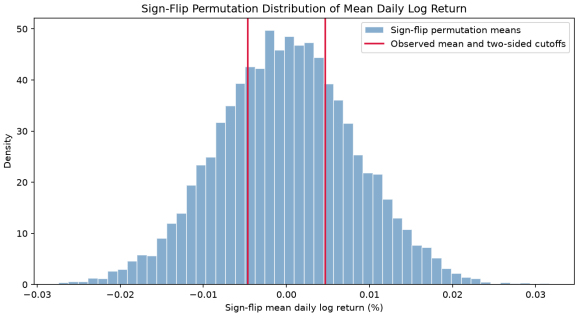
Hypothesis Tests: No Evidence of Non-Zero Mean

Test	Statistic	p-value
Wald z	-0.565826	0.571512
Sign-flip	—	0.570343

Decision at $\alpha = 0.05$

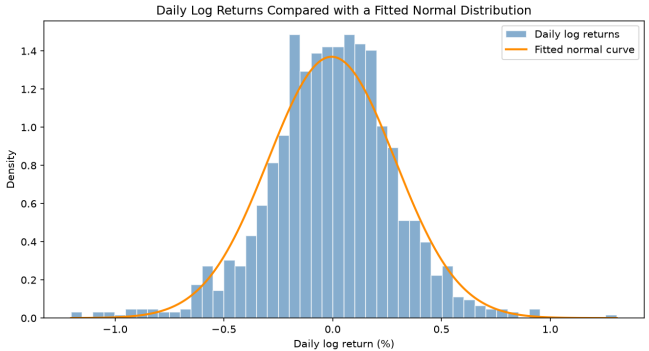
Both tests fail to reject H_0.

The data do not provide strong evidence that the mean return differs from zero. This does not prove that the true mean is exactly zero.



Sign-flip null distribution with observed two-sided cutoffs.

Return Distribution: A Useful Caution



Observed returns compared with a normal curve fitted by sample mean and SD.

- Sample skewness: -0.253.
- Excess kurtosis: 1.362.
- The shape is not perfectly normal, especially in the tails.
- Large-sample inference may still be useful, but dependence and volatility clustering remain concerns.

What Should a Decision-Maker Take Away?

Main finding

Across the tested methods, the estimated average return is small relative to its uncertainty, and the evidence does not indicate a non-zero mean over this sample.

- The agreement between interval estimates and tests is reassuring for this particular sample and estimand.
- A near-zero estimated mean does not imply low exchange-rate risk: daily variation remains substantial.
- Risk decisions should consider volatility, exposure, and the investment or planning horizon, not the mean test alone.

Limitations

- **Observation intervals:** Missing source values mean some returns span more than one calendar day.
- **Dependence:** The Wald standard error treats observations as independent; FX returns can show volatility clustering.
- **Permutation assumption:** Sign-flipping relies on symmetry under the null.
- **Scope:** One exchange-rate series and one sample window; results may differ across regimes or currency pairs.
- **Reproducibility:** The processed CSV is provided, but the workbook-to-CSV cleaning transformation is not yet scripted.

Possible next step

Compare with a block bootstrap or dependence-robust inference, and test whether conclusions change across sub-periods.

Conclusion

Different methods, same conclusion:

we do not reject a zero mean return at the 5% level.

Thank you.

Questions?